

RecSys challenge 2022 Innovation

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1 Introduction

The introduction will cover a more detailed background for our innovation of the paper: *Leveraging Ensembles of Recommenders to Compete with Budget Constrained Resources*. For a comprehensive understanding we recommend reviewing both our innovation proposal and the anchor paper prior to proceeding. We chose the innovation of a marketing-driven optimization strategy. The innovation includes boosting performance of an item subgroup chosen by item feature values. We thought of 4 possible directions for implementation:

- **Update Loss Function:** Modify the loss function for either candidate selectors and/or rankers to better handle the item subgroup.
- **Augment Additional Data:** Extend sessions that result in purchases from the subgroup.
- **Add Model with Different Task:** Introduce a new model for prediction or feature enhancement, such as a binary classifier to predict purchase likelihood from item subgroup given a session. This could then add a flag feature used in the pipeline.
- **Feature Engineering:** Incorporate new features like co-occurrence or TF-IDF related to the item subgroup.

We decided to focus on updating the loss function and augmenting data. Specifically, we adjusted the lightGBM ranker loss function to use sample weights, giving sessions leading to subgroup item purchases a weight of $1 + \alpha$ ($\alpha = 0.5$). This is expected to boost performance on ranking candidates from the subgroup while maintaining generalization for the entire task.

Regarding data augmentation, we observed that only 73.2% of sessions had the correct item among the list of candidates for train data. This lead us to attempt to improve Recall, particularly for sessions with a subgroup label. To achieve this, we performed the entire cycle of data preprocessing and retraining the model using both the original and augmented data sets. We initially considered using GANs, like CTGAN (Conditional Tabular GAN), but chose a simpler method for augmenting the data. We selected sessions leading to subgroup item purchases and applied a Bernoulli trial with $p = 0.5$ to decide whether to keep or remove each view. Sessions were then filtered to ensure at least one view remained, excluding unchanged sessions. This process generated new unique sessions labeled with the subgroup. It is important to note that the removal of viewed items has its implications on session derived features, for example session time length may be affected, leading to misleading features.

For the subgroup selection process, we performed a random methodology to choose category and feature value pairs, aggregating them into a set until they represented at least 10% of the total items. We saved these values for use of item subgroup extraction along the pipeline. This global approach allows for more flexibility, rather than selecting specific item IDs upfront. For example if we want to promote winter items, this allows for future items with this feature to be streamlined in the pipeline. The outcome of this random selection process led to selecting a feature set with 12.5% of items falling within our defined subgroup. Training item occurrences consisted of 9.5% of the subgroup, and 7.6% of test items occurrences. Approximately 6% of test sessions had a label from the subgroup, making for a large enough subgroup for evaluating our approach.

2 Challenges

Incorporating modifications into existing frameworks was challenging. Implementing our innovative approach necessitated the integration and modification of existing code within notebooks. The complexity of this task was

significantly amplified by the initial poor readability of the codebase, presenting a greater challenge compared to developing new code from scratch.

The complex execution environment made running multiple experiments difficult. The execution of our models involved navigating a complicated structure of directories, each containing unique data sets, candidate models, and configuration settings. This complexity made the process of compiling final results for comparative analysis more complicated than a straightforward execution of a singular notebook.

3 Code

The project codebase is placed in a shared google drive folder. To view the execution process or replicate the results, you can either explore the provided run notebooks in this *Link* or execute the notebooks using the original dataset. All relevant notebooks and newly processed data are distinguished by an 'innovation' tag for easy identification.

Data Preparation: The 'augment_traindata' notebook, located under the data_preparation directory, details the processes of data augmentation and subgroup selection. Additionally, within the 'Innovation Part' subdirectory, you can find notebooks tailored for data preparation, where processed data is stored separately for further use.

Models for Ensemble: For candidate selectors, refer to the 'Innovation Part' within the 'Models for ensemble' directory. This section contains specific notebooks dedicated to modeling components of our innovative approach.

Final Submission: To review the final outcomes, navigate to the 'Final_submission' directory, where the 'Innovation' file compiles all notebooks presenting the final results.

4 Experiments

In our study, we conducted an offline comparative analysis to assess the impact of various innovation components on our recommendation system. We used the original ensemble (from anchor paper) as our baseline for comparison. The modifications we evaluated included an updated loss function and the incorporation of additional augmented data into the updated pipeline. A key focus of our analysis was to compare the overall MRR@100 performance against the subgroup-specific MRR@100. This comparison aimed to determine whether enhancements targeting the subgroup could potentially compromise the system's overall performance. Additionally, we examined the effect of simplifying the candidate selection process by limiting it to using only the UCF and GRU methods. This was intended to evaluate the system's efficiency and effectiveness in a constrained scenario. Our analysis extended to both training and validation datasets to test the model's ability to generalize on unseen data.

5 Results

Table 1 presents a comparative analysis of MRR@100 scores across various configurations of our new ensemble model. All models used URM_W as the user rating matrix, where 'ALL' includes UCF, ICF, ICBF and GRAPH as candidate selectors. The setup used 10 lightGBM rankers with 100 estimators after candidate selection. Updated loss function is referred to as 'Loss' and the incorporation of additional augmented data is referred to as 'Loss+Augmented'. Test is referred to as the validation set which in our setup is the leaderboard dataset.

Table 1: MRR@100 Results

	Baseline		Loss		Loss+Augmented	
	UCF+GRU	ALL+GRU	UCF+GRU	ALL+GRU	UCF+GRU	ALL+GRU
Train	0.1734	0.1866	0.1735	0.1868	0.1567	0.1728
Train Subgroup	0.2066	0.2157	0.2068	0.2184	0.1878	0.2027
Test	0.1919	0.1918	0.1920	0.1908	0.1899	0.1897
Test Subgroup	0.2005	0.2022	0.2000	0.2016	0.2139	0.2189

6 Insights

Upon examining the impact of changing the loss function on the outcomes, we observed that there was a slight improvement in the training data performance, along with a slight decline in the test data performance. A plausible explanation for this phenomenon could be overfitting, potentially caused by an overly large alpha parameter. This parameter significantly affects the weighting of the loss function for the item subgroup.

When examining the effects of combining augmentation with the modified loss function, we observed distinct outcomes. As anticipated, there was a decrease in training performance, which can be attributed to the inclusion of additional data to the original train data. However, we noted a significant 8.2% increase in test MRR for the specific subgroup, while the overall MRR was only reduced by 1.1%. This suggests that employing the augmentation method in conjunction with a weighted loss function yielded much better results than merely altering the loss function alone.

7 Project Extension

- **Additional enhancements:** Exploring our additional proposed enhancements beyond the implementation of updating the loss function and augmenting data. Future work could benefit from integrating feature engineering techniques and adding specialized models. This could build on the existing methodology.
- **Optimization of hyper-parameters:** Searching for optimal alpha could prove that updating loss function alone may be a valid approach.
- **Cross validation:** Conducting iterative experiments with varying sets of feature values is crucial to assess the robustness of the model. By selecting different feature values for each experiment, we can evaluate if the improvements observed with the initial set of feature values hold consistent across different subgroups. This testing will help ensure that the model's enhanced performance is not contingent on a specific set of feature values but is broadly applicable across various item characteristics.
- **Isolate data augmentation:** Run additional experiment for evaluating only data augmentation, this could help isolate the results and pinpoint the exact effects of the data augmentation component.